

UNDERSTANDING DOCKER NETWORKING

❖ Introduction to Docker Network

- Docker networking enables containers to **communicate with each other**, the host system, and external networks. It allows containers to be isolated or connected based on the chosen network type.
- **Why is Docker Networking Important?**
 - Containers need to communicate for distributed applications.
 - Enables service discovery between containers.
 - Offers flexibility in container-to-container or container-to-host communication.
 - Supports network isolation for security and performance.

❖ Types of Docker Network Drivers

Docker provides several built-in network drivers that define how containers communicate:

Driver	Description
bridge	Default network driver for standalone containers. Provides private internal networking on the host
host	Shares the host's network stack. No isolation between container and host network.
none	Disables networking for the container.
overlay	Used in Docker Swarm mode to connect containers across multiple hosts.
macvlan	Assigns a MAC address to the container, allowing it to appear as a physical device on the network. Used for advanced networking.

❖ Docker Network Commands

Command	Description
docker network ls	List all available networks
docker network inspect <name>	View detailed info about a network
docker network create <name>	Create a new custom network
docker network rm <name>	Delete a custom network

❖ Creating and Deleting Networks

- **Create a Custom Network**
`docker network create my_custom_net`
- **Create a Custom Network with Specific Driver**
`docker network create --driver bridge my_bridge_net`
- **Delete a Network**
`docker network rm my_custom_net`